NIHVA
NIHVA Technologies Pvt. Ltd.

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DOCK-LEVELER

OPERATION & MAINTENANCE MANUAL

1) DOCK LEVELER – NDL-091
2) DOCK LEVELER – NDL-092

DOCK LEVELER	MANUAL

A DESPATCH & UNLOADING

1. Loading, for dispatch, will be our responsibility. On receipt of the Dock-leveler at site, please check and ensure that there is no physical damage to the equipment.

2. While unloading the equipment from the transporting vehicle, please ensure to use slings with rated capacity higher than the gross weight of the Stationary lift in packed condition.

3. Slings MUST be hooked ONLY at specified positions. These positions are indicated and marked with a figure of 'HOOK'. Improper sling position or improper slinging could cause damage to the equipment.

B. ORDERING OF SPARES

1. When ordering for spare, please specify the following in your Purchase Order.

   - Equipment Name
   - Model Number
   - Equipment Serial No.
   - Part name / Description
   - Part Number
   - Quantity required
   - Delivery period / time
   - Mode of dispatch

   This would avoid ambiguity and help us serve you better.

DATASHEET FOR DOCK LEVELER – NDL-091

CLIENT: - TAKENAKA INDIA, GUJRAT

1.0 CONSTRUCTION DETAILS

1.1 Required Capacity - each	: 06 ton
1.2 Model No.	: NDL-91
1.3 Leveler operation mode operated	: AC power pack hydraulically
1.4 Lip operation mode	:hydraulic
1.5 Operation control	: by push buttons and control panel
1.6 Loading platform height from road Level	: 1200 mm (Will be confirmed during Detailing)
1.7 Travel above & below dock level	: (+) 350 / (-) 250 mm
1.8 Platform Size (w x l)	: 2200 x 2850 mm
1.9 Lip size (w x l)	: 2200 x 400 mm
1.10 Working length	: 3250 mm
1.11 Overall working size	: 2200x3250 mm
1.12 Maximum slope can be achieved	: 8 to 10%
1.13 Clear Pit dimensions (W x L x D)	: 2250x 2910 x 485 mm
1.14 Hydraulic power pack	
1.14.1 Make	"NIHVA"
1.14.2 Driver type	: Direct by electric Motor
1.14.3 Absorbed power	: 0.75 kW
1.14.4 Motor Ratings	: 0.75/1 kW /HP
1.14.5 Degree of protection for motor: IP 55	
1.14.6 Class of Insulation for motor	: Class F with temp. Rise limited to Class B
1.14.7 Motor RPM	: 1410
1.14.8 Pump design flow	: 1.8 LPM
1.14.9 Pump pressure	: 200 (max) Kg/cm²g
1.14.10 Safety valve set pressure	: 90 Kg/cm²g
1.14.11 in built safety valve	: Yes
1.14.12 Pump type	: Gear
1.14.13 Valves required	: By vendor
1.14.14 Volume of oil receiver	: 5 Lit
1.14.15 Dimensions of power pack Unit	: 400 H x 560 L x 340 W mm
1.14.16 First fill of oil / Qty	: By Vendor
1.14.17 Hydraulic tubing	: By Vendor
1.14.18 Instrumentation	: By vendor
1.14.19 Control panel	: By vendor

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DATASHEET FOR DOCK LEVELER – NDL-092

CLIENT: - TAKENAKA INDIA, GUJRAT

1.0 CONSTRUCTION DETAILS

1.1 Required Capacity - each	: 06 ton
1.2 Model No.	: NDL-92
1.3 Leveler operation mode operated	: AC power pack hydraulically
1.4 Lip operation mode	:hydraulic
1.5 Operation control	: by push buttons and control panel
1.6 Loading platform height from road Level	: 1200 mm (Will be confirmed during Detailing)
1.7 Travel above & below dock level	: (+) 350 / (-) 250 mm
1.8 Platform Size (w x l)	: 2200 x 2925 mm
1.9 Lip size (w x l)	: 2200 x 400 mm
1.10 Working length	: 3325 mm
1.11 Overall working size	: 2200 x 3325 mm
1.12 Maximum slope can be achieved	: 8 to 10%
1.13 Clear Pit dimensions (W x L x D)	: 2250x 2985 x 485 mm
1.14 Hydraulic power pack	
1.14.1 Make	"NIHVA"
1.14.2 Driver type	: Direct by electric Motor
1.14.3 Absorbed power	: 0.75 kW
1.14.4 Motor Ratings	: 0.75/1 kW /HP
1.14.5 Degree of protection for motor: IP 55	
1.14.6 Class of Insulation for motor	: Class F with temp. Rise limited to Class B
1.14.7 Motor RPM	: 1410
1.14.8 Pump design flow	: 1.8 LPM
1.14.9 Pump pressure	: 200 (max) Kg/cm²g
1.14.10 Safety valve set pressure	: 90 Kg/cm²g
1.14.11 in built safety valve	: Yes
1.14.12 Pump type	: Gear
1.14.13 Valves required	: By vendor
1.14.14 Volume of oil receiver	: 5 Lit
1.14.15 Dimensions of power pack Unit	: 400 H x 560 L x 340 W mm
1.14.16 First fill of oil / Qty	: By Vendor
1.14.17 Hydraulic tubing	: By Vendor
1.14.18 Instrumentation	: By vendor
1.14.19 Control panel	: By vendor

DOCK LEVELLER

INTRODUCTION:-

The hydraulic Dock Leveler is the result of many years' experience in materials handling engineering. "NIHVA " offers the best possible factor of safety with simple, easy to use controls. The high quality of manufacture gives a robust construction and many years of useful life. Each lip and platform movement can be individually controlled (or corrected) by push button.

The dock leveler can be set at any position. During loading, the Dock Leveler is in floating mode i.e. it follows the up/down movements of the vehicle deck. Even with lopsided vehicles, the Dock leveler lip will remain in contact, and toe-traps are avoided by the tilt built into the Dock Leveler.

The whole platform is designed to flex laterally adequately to adapt to sideways tilted vehicles. The leveler always seals perfectly to the lorry bed.

PRINCIPLE:-

Dock Leveler is an adjustable platform used between the transport vehicle and dispatch area of the warehouse of factory for efficient and safe material transfer applications. It bridges the gap between the dispatch area and the transport vehicle and compensates for variations in the truck bed height.

CONSTRUCTION:-

It basically consists of heavy duty fabricated Top Chequered platform and bottom angle channel construction frame. Both are joined by stainless steel machined pins. Lip is hinged to the top platform by means of stainless steel pins.

Top Platform is reinforced with I Beam welded below the platform to strengthen the platform. The cylinder is mounted at the best advantage point against the top platform and bottom frame. Compact power pack includes a single HP motor with overload protection. Hydraulic Pump is Dowty make. Hydraulic operations are by using solenoid control valve.

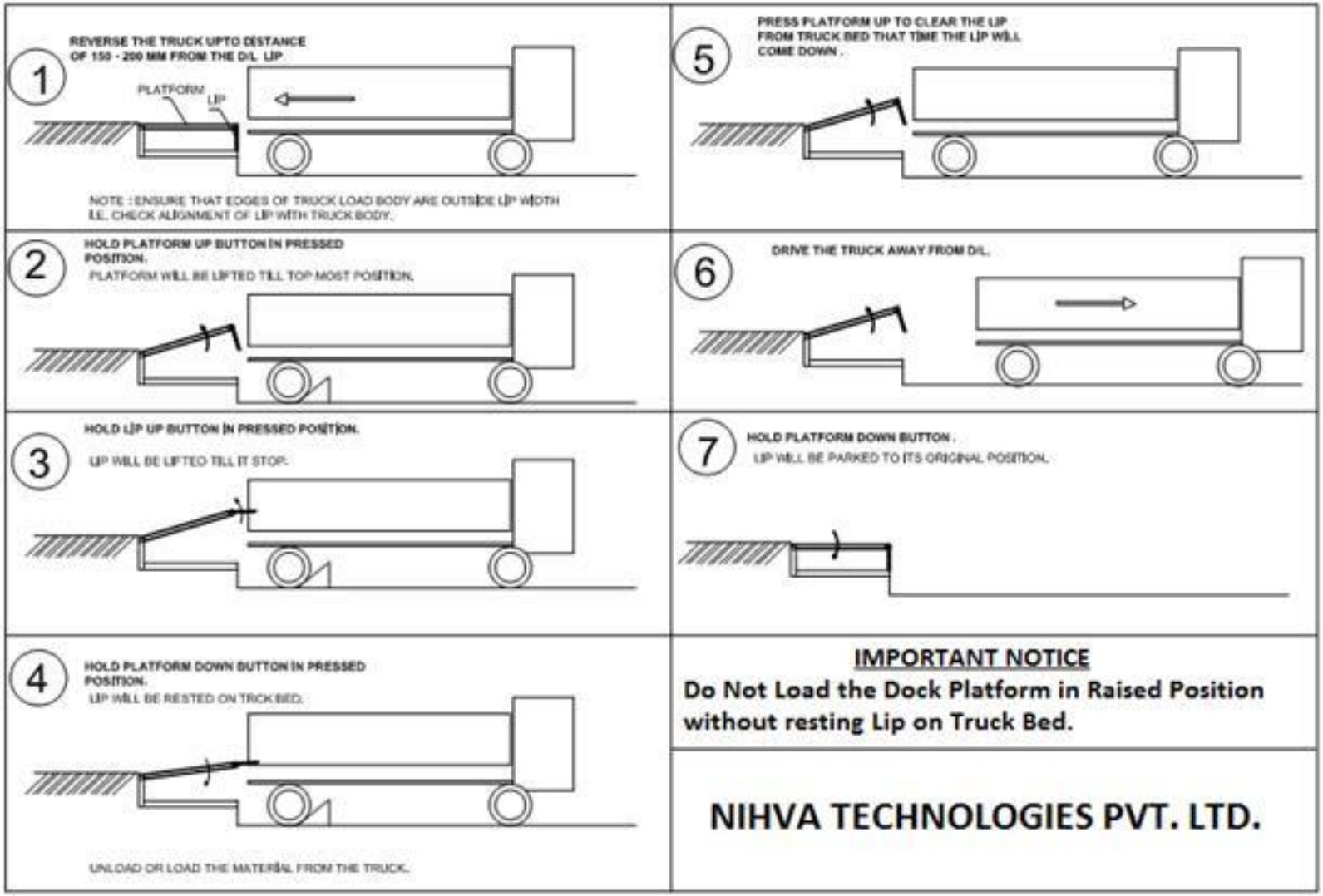
FEATURES:-

1. Convenient location of all controls.
2. Independent deck and lip operation.
3. Controls for quicker cycling time.
4. Smooth dependable hydraulics.
5. Quite operation at the touch of the button.
6. Lower operating cost and fewer maintenance requirements.
7. Recommended for container & loading unloading applications.
8. Maintenance Strut.
9. Toe guards.
10. Heavy thickness of the lip gives good life.
11. Self standing structure transfers load to the base of the pit, independent of pit skirting civil work.
12. Profiled hinges gives good welding strength.
13. No cross support imparts good flexibility.

PRE – OPERATING INSTRUCTIONS:-

1. Ensure the dock platform is empty.
2. Ensure the truck is parked properly.
3. Ensure the chocks to the truck to arrest the movement while loading.
4. Ensure proper functioning of the switches.
5. Ensure the truck height is within the range.
6. Ensure no joint leakage.

DOCK LEVELER MANUAL



ROUTINE MAINTENANCE AND LUBRICATION

These recommendations are applicable to the Dock Leveler considering average daily usage of 15 to 20 times. If it is used more frequently, the same time intervals for the maintenance should be adjusted accordingly.

DAILY:-

1. Check Hydraulic pipe lines for leak proof working.
2. Check electrical connections for their proper working.

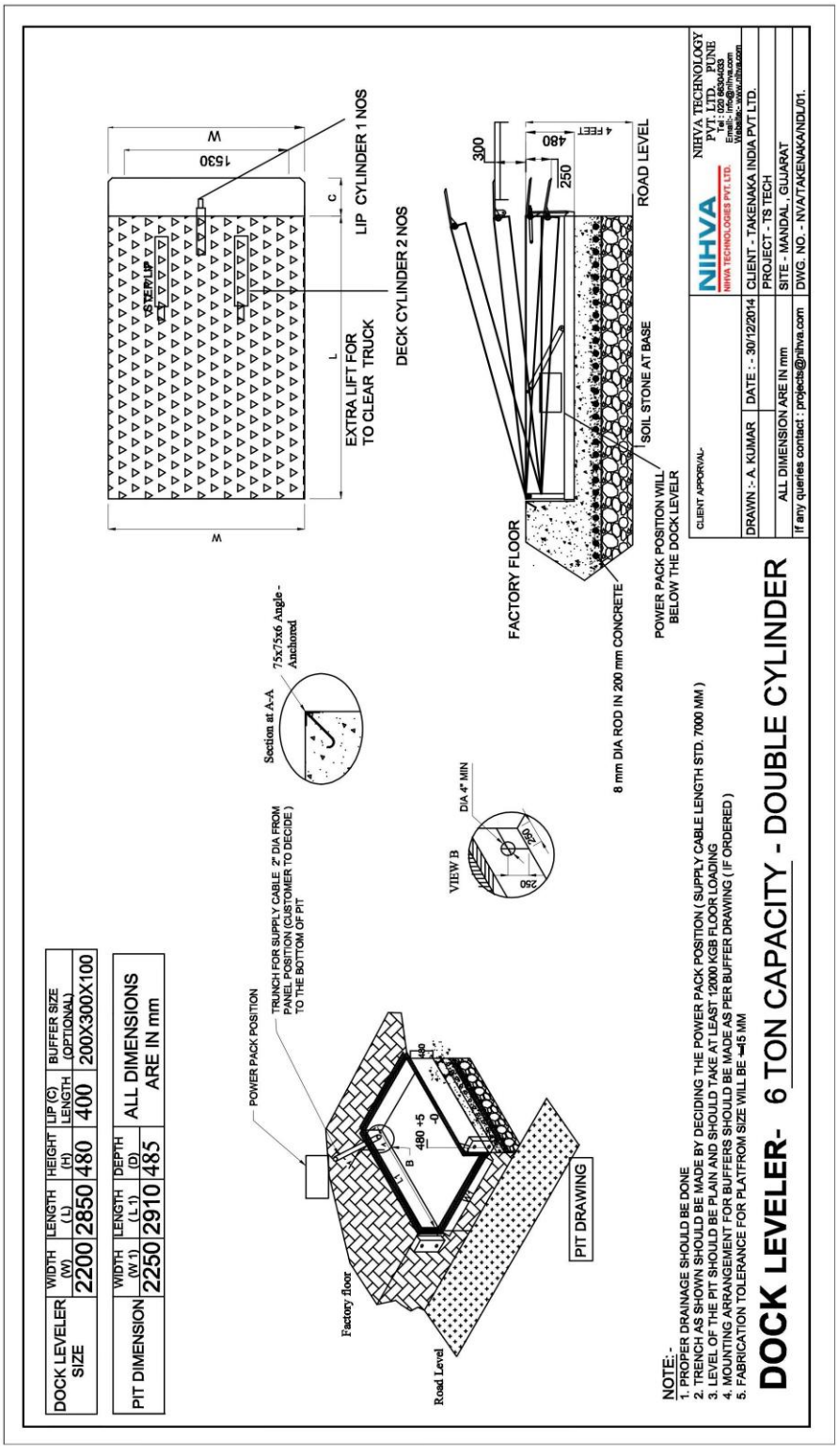
MONTHLY:-

1. Grease the hinge pins at bottom frame and top plate joint and extension lip plate hinge joint.
2. Grease the main Jack Cylinder and Lip Jack Cylinder joints.

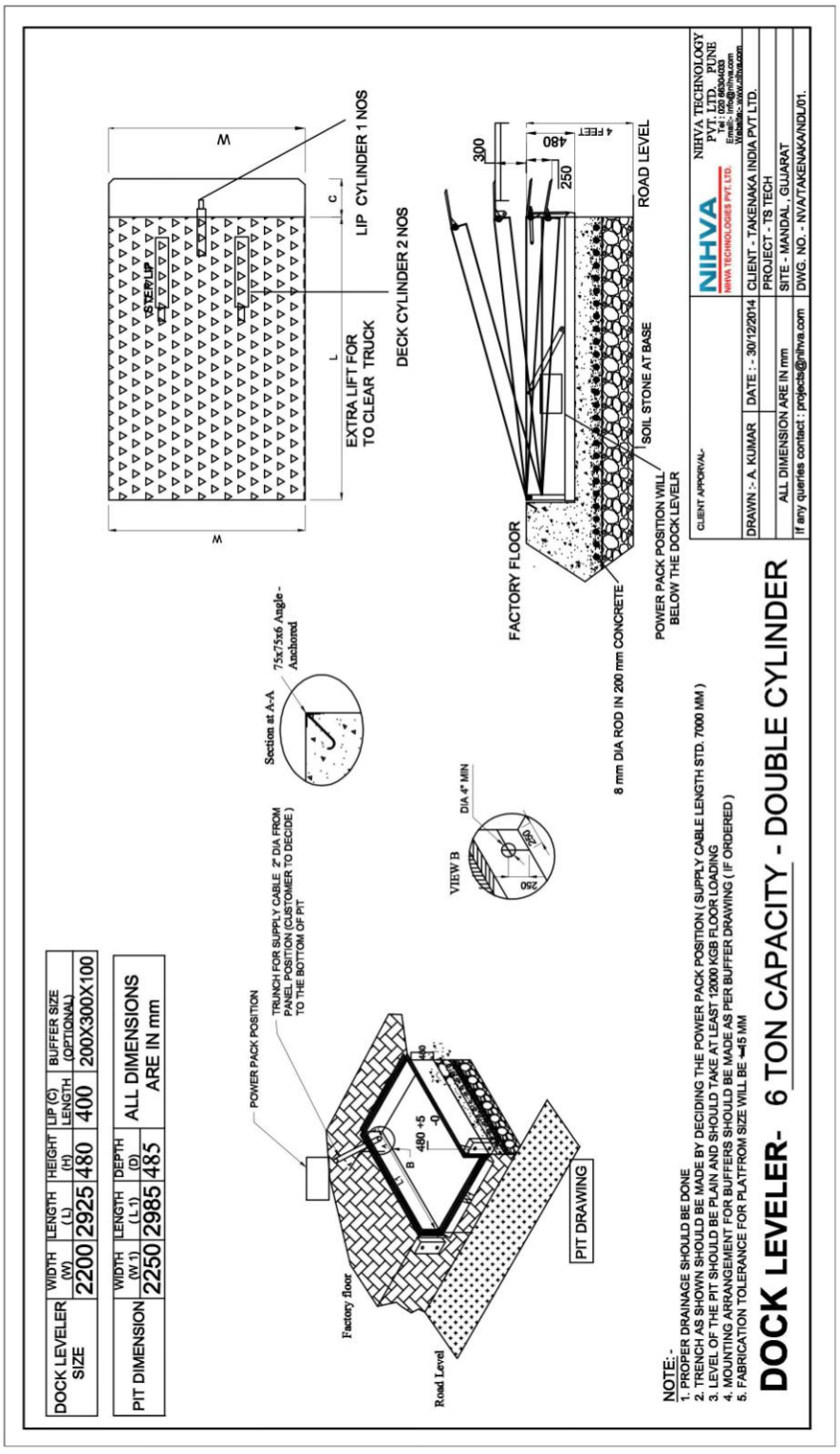
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ASSEMBLIES
&
SPARE PARTS
LIST

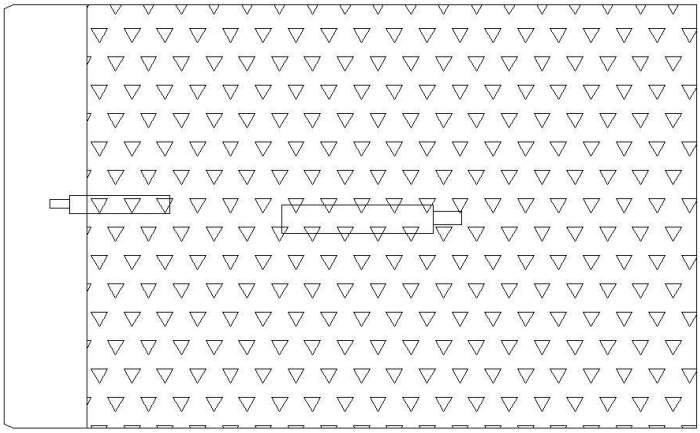
DOCK LEVELER	MANUAL

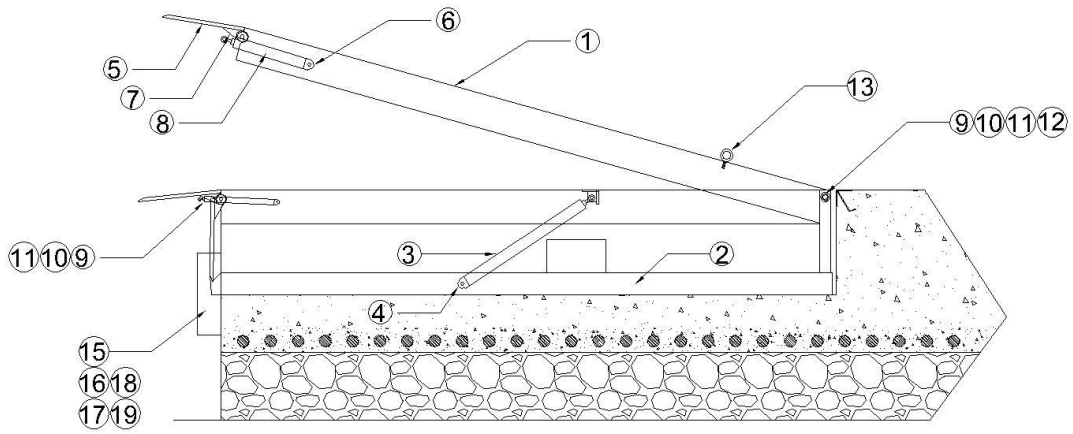


DOCK LEVELER MANUAL



DOCK LEVELER MANUAL



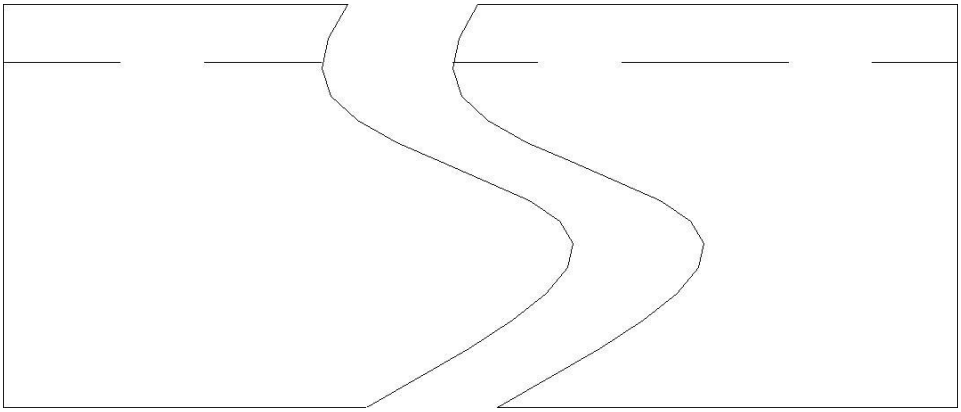


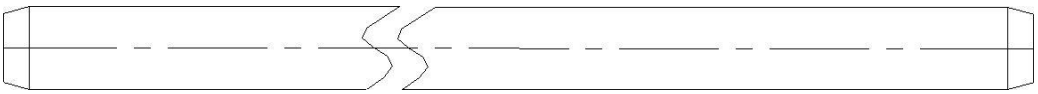
DOCK LEVELER

MANUAL

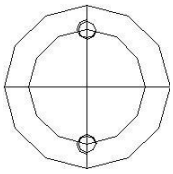
SR. NO.	DESCRIPTION	QTY.
1	TOP PLAT FORM WELDED ASSLY	1
2	BOTTOM FRAME WELDED ASSLY	1
3	MAIN JACK ASSLY	2
4	MAIN JACK PIVOT PIN	4
5	LIP PLATE	1
6	LIP JACK BOTTOM PIVOT PI N	1
7	LIP JACK TOP PIVOT PIN	1
8	LIP JACK ASSLY	1
9	PLATFORM & LIP PIVOT PIN	2
10	PIN LOCKING WASHER	2
11	M6 X 20 L CSK BOLT	8
12	GREASE NIPPLE	6
13	I BOLT M16	4
14	ANCHOR FASTNER	8
15	RUBBER BUFFER	2
16	RUBBER BUFFER PIVOT PIN	4
17	SPLIT PIN	8
18	RUBBER BUFFER MTG. ANGLE	4
19	ANCHOR FASTNER	8

SPLIT LIP PLATE
PART NO-70000160
QTY-01

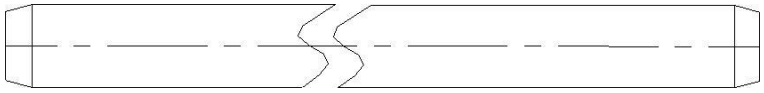




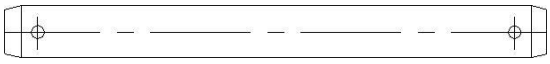
PLATFORM PIVOT PIN (REAR)
PART NO-70000161
QTY-01



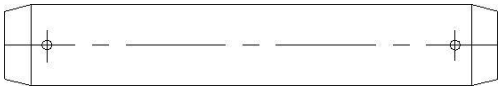
PIN LOCKING WASHER
PART NO-70000164
QTY-04



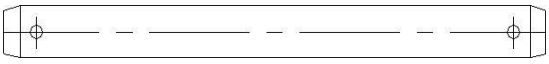
LIP PLATE PIVOT PIN (M.S)
PART NO-70000162
QTY-01



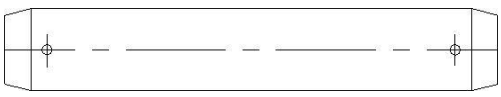
LIP JACK PIVOT PIN(TOP)
PART NO-70000167 QTY-
01



MAIN JACK PIVOT PIN(TOP)
PART NO-70000165
QTY-01



LIP JACK PIVOT PIN(BOTTOM)
PART NO-70000168
QTY-01



MAIN JACK PIVOT PIN(BOTTOM)
PART NO-70000166
QTY-01

DOCK LEVELER MANUAL

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SR.NO	ITEM NO.	PARTS NO.	DESCRIPTION	SIZE	QTY
15	15	10000071	SINTERED BUSH	1"X1.1/8"X1"	2
14	14	10000070	ROD END BEARING	M24X2XPH25PIN	1
13	13	WPS00015	WIPER SEAL	Ø40xØ48x7	1
12	12	PTS00011	PISTON U SEAL	Ø50xØ40x9	1
11	11	ORN00040	O RING	Ø48X2.5	1
10	10	LKW13225	LOCKING WIRE	13 GAUGE	1
9	9	CP00009	CIRCLIP	EXT Ø25	1
8	8	70000050	SPACER Ø40	Ø40XØ49XSP	1
7	7	70000154	PISTON Ø40	Ø50XØ40X52	1
6	6	70000049	WAHSER RETAINER SEAL	Ø48XØ25.2X5	1
5	5	70000048	WIPER SEAL HOUSING	Ø63X82 L	1
4	4	70000047	WASHER	Ø49XØ32X4	1
3	3	70000159	BOTTOM END	Ø63X49	1
2	2	70000158	ASSLY WELDED CYLINDER	2" HONEDx	1
1	1	70000157	SHAFT JACK	Ø40 x 580	1

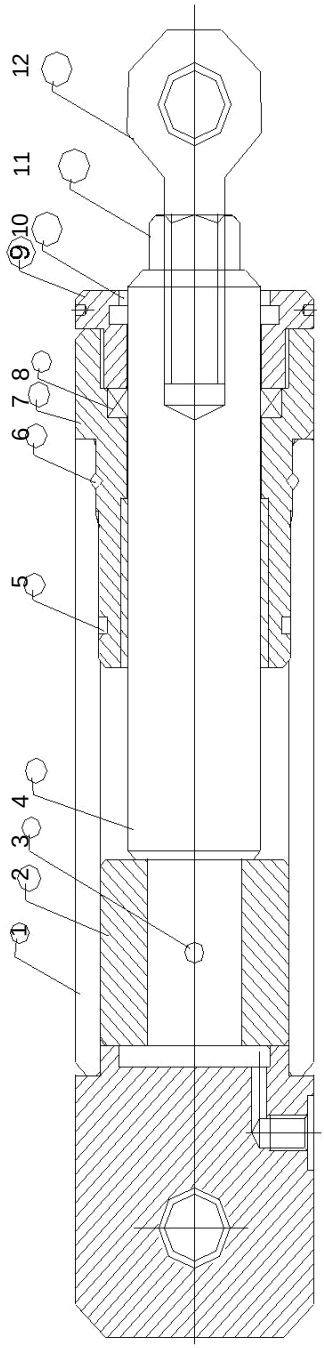
MAIN JACK ASSLY

PART NO-70000156

DOCK LEVELER

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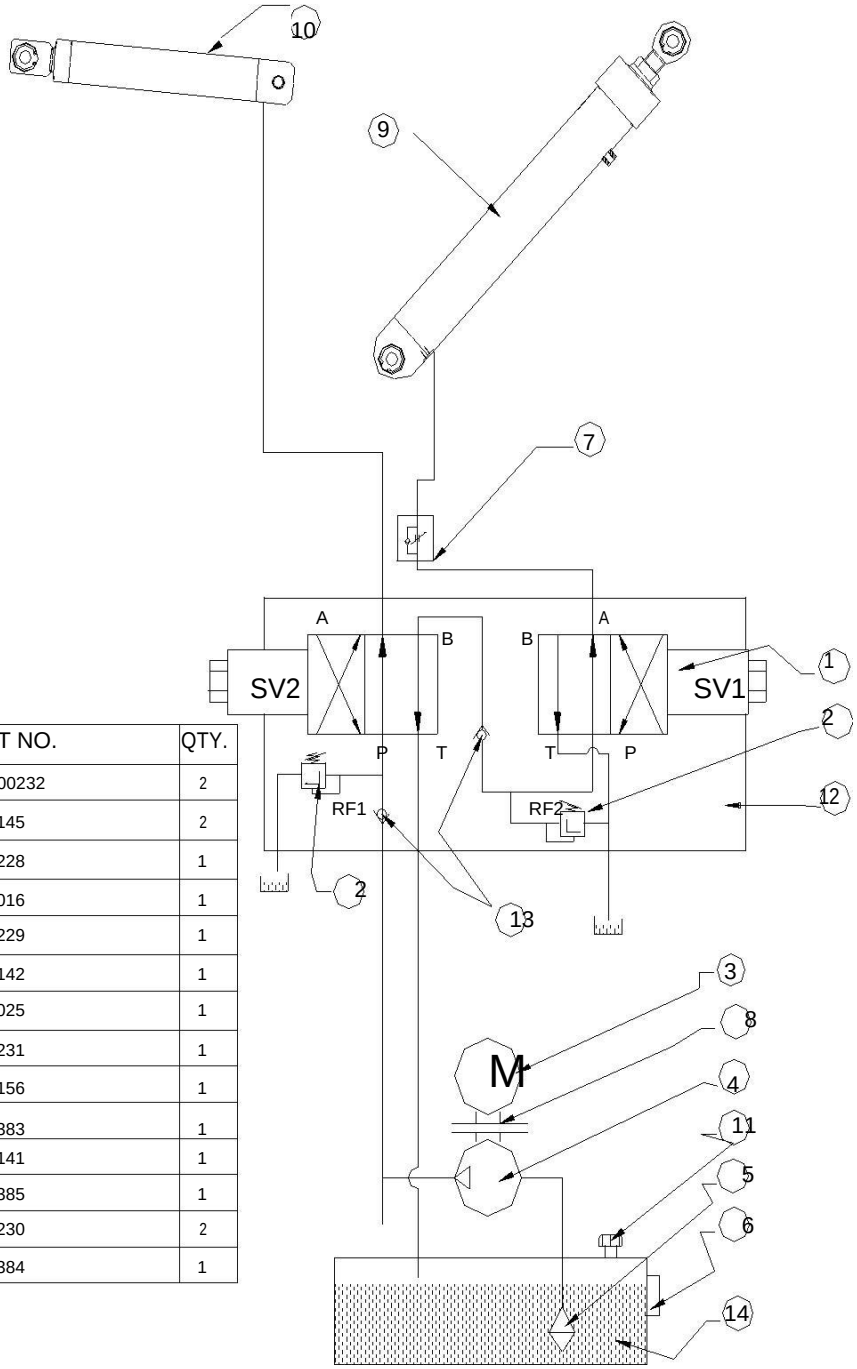
SR.NO.	PARTS NO.	DESCRIPTION	QTY
12	REB00006	ROD END BEARING	1
11		HEX NUT M-16X2P	1
10	WPS00002	SEAL WIPER	1
9	70000390	WIPER SEAL HOUSING	1
8	RDS00003	SEAL ROD	1
7	70000389	SEAL HOUSING	1
6	LKW13235	WIRE LOCKING	1
5	ORN00002	O RING	1
4	70000388	SHAFT(PLATED)	1
3	DSP00001	SPRING PIN	1
2	70000387	PISTON	1
1	70000386	JACK CYLINDER	1

LIP JACK ASSLY

PART NO-70000383

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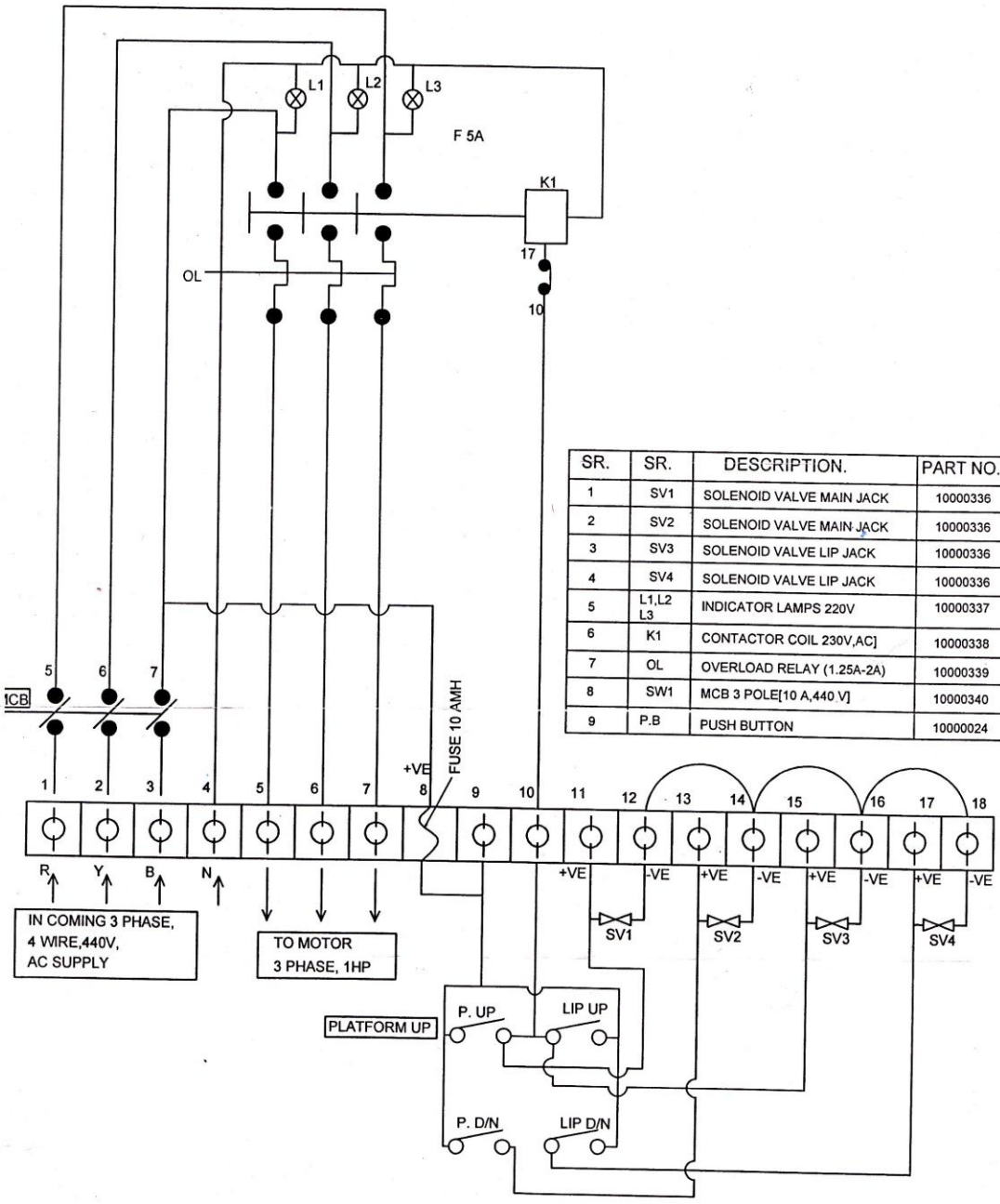
HYDRAULIC CIRCUIT FOR DOCK LEVELER



SR.NO.	DESCRIPTION.	PART NO.	QTY.
1	SOLENOID VALVE (SV1,SV2)	10000232	2
2	RELIEF VALVE (RF1,RF2)	10000145	2
3	MOTOR 1HP,3PHASE	10000228	1
4	GEAR PUMP(OP-3006)	10000016	1
5	SUCTION STRANER	10000229	1
6	OIL LEVEL INDICATOR	10000142	1
7	THROTTLE VALVE	10000025	1
8	LOVEJOY COUPLING	10000231	1
9	MAIN JACK ASSLY.	70000156	1
10	LIP JACK ASSLY.	70000383	1
11	BREATHER FSB-25	10000141	1
12	MANIFOLD BLOCK	70000385	1
13	IN LINE CHECK VALVE 1/4"	10000230	2
14	OIL TANK 15LTR CAPACITY	70000384	1

DOCK LEVELER | **MANUAL**

DOCK LEVLER 4SOLENOID VALVE AC PANEL



SR.	SR.	DESCRIPTION.	PART NO.	QTY.
1	SV1	SOLENOID VALVE MAIN JACK	10000336	1
2	SV2	SOLENOID VALVE MAIN JACK	10000336	1
3	SV3	SOLENOID VALVE LIP JACK	10000336	1
4	SV4	SOLENOID VALVE LIP JACK	10000336	1
5	L1,L2 L3	INDICATOR LAMPS 220V	10000337	3
6	K1	CONTACTOR COIL 230V,AC]	10000338	1
7	OL	OVERLOAD RELAY (1.25A-2A)	10000339	1
8	SW1	MCB 3 POLE[10 A,440 V]	10000340	1
9	P.B	PUSH BUTTON	10000024	1

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